GIS for ICT Procurement by Government Agencies

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CS 191 WFQ1

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I. Web App Requirements Model

a. Content Model

Figure 1a.1 below shows that there are five main components to the ICT Procurement GIS web app: the map itself, search filters, purchase details, location comparison, and data status. A data tree connecting all components would be too large to fit into one page, so we would be splitting them up into subparts.

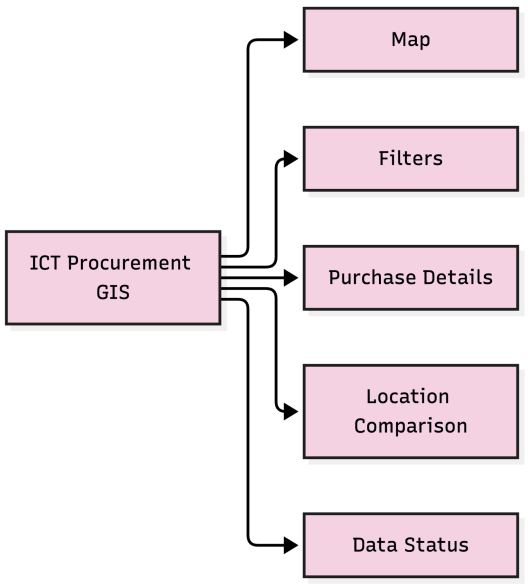


Figure 1a.1: Main components of the web app

Figure 1a.2 shows that under the map component, we have two main subcomponents: the zoom controls for the user to change the zoom level of the map to show more or less area of a location, and location selectors for government agencies that have matching result with the filters to be applied when searching for items.

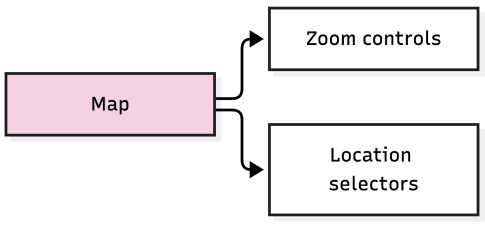


Figure 1a.2: Subcomponents of the map component

Figure 1a.3 displays the subcomponents under the filters component. We can filter data using the following: the item search bar for searching by using keywords, the price range slider, selection of specific regions, provinces, and cities, and the specific year of granting the award notice.

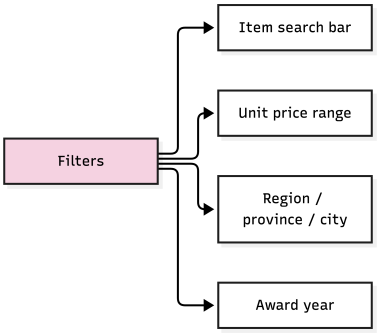


Figure 1a.3: Subcomponents of the filter component

Figure 1a.4 shows the subcomponents under the purchase details component. In viewing the details of a purchase transaction, we show the government agency and its location, the procurement notice title, the bid reference number, award date, supplier, contract price, item description, quantity, unit price, and a link to the PhilGEPS portal for cross-checking and verification.

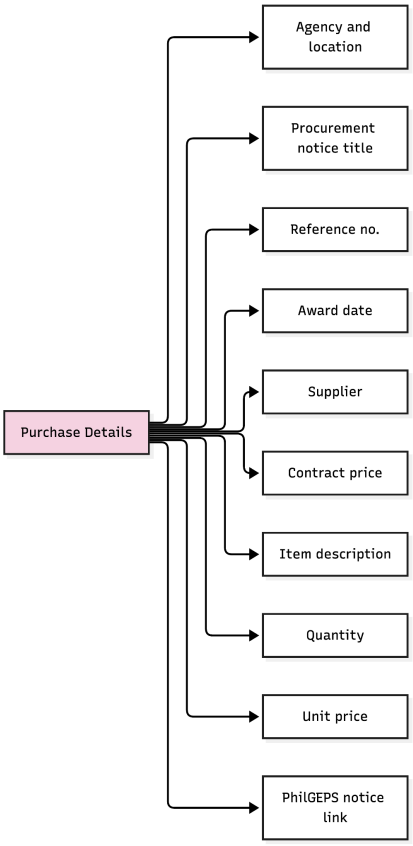


Figure 1a.4: Subcomponents of the purchase details component

Figure 1a.5 displays the location comparison component, with its subcomponents which are the selected item and its price at a selected agency, price compared with another agency's or an average, the average level being either some region or for the whole country, and the price difference.

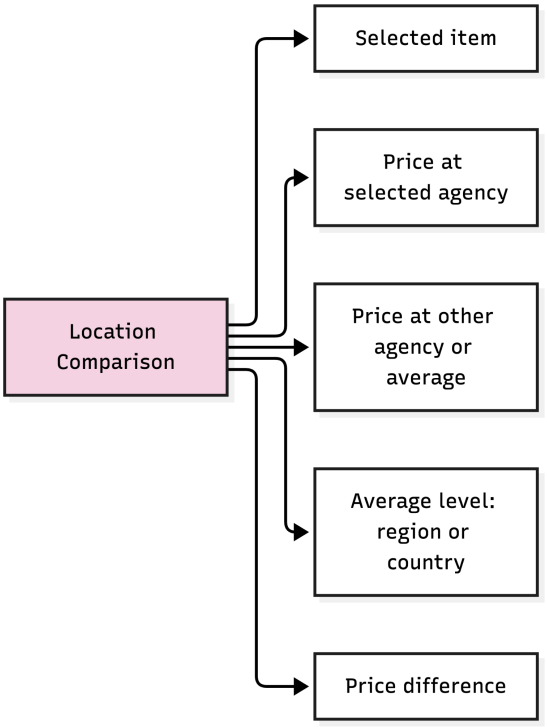


Figure 1a.5: Subcomponents of the location comparison component

Figure 1a.6 shows the subcomponents under the data status component. The last updated date shows when the site was last updated with new data, since PhilGEPS uploads new datasets every three months. The datasets included subcomponent is necessary to show which datasets are included in the current version of the site, since as a proof of concept, we would only initially work with data from NCR in the past three months, with the other datasets to follow in the upcoming phases of this project.

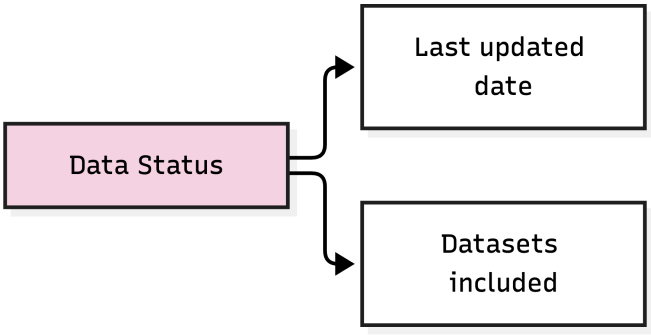


Figure 1a.6: Subcomponents of the data status component

b. Interaction Model

i. Actors and Goals

Actor	Type	Goal
Administrator	Initiating	Keeps the data current by importing updated PhilGEPS datasets and checking that the import worked.
User (procurement officer, auditor, supplier, researcher, citizen)	Initiating	See what government agencies paid for an ICT item, compare prices across places, and check any price against the original PhilGEPS notice.

For this project, we have two main actors. The first actor is the **administrator**, which keeps the site updated by importing the latest datasets. The second is the **user**, which can be any person who accesses the website for various purposes. For example, a procurement officer looks up what other agencies paid for similar ICT items before setting a budget. An auditor checks whether a price looks unusually high compared with similar purchases. A supplier sees what is being paid and where, so they can price competitively. A researcher studies government ICT spending without collecting the data by hand. Any citizen can even access the site just to see how the government is spending their taxes. No account is needed, and all of them do the same things in the system: search for an item, filter and view prices on the map, compare locations, and open a purchase to verify it against the original notice.

ii. Use Cases

The Administrator actor connects only to “Import PhilGEPS data,” as shown in Figure 1b.1 below. In this use case, the admin loads completed ICT projects from the PhilGEPS at regular intervals (about every 3 months), cleans and stores the records, and updates the data status.

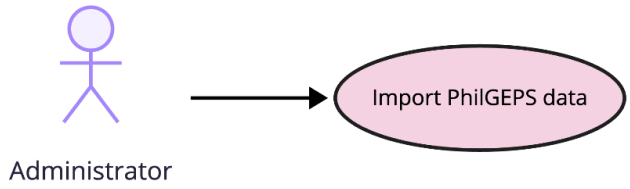


Figure 1b.1: Use case diagram for the Administrator actor

The user, as shown in Figure 1b.2, can search for an ICT item, view the price map, view location purchases, compare locations, view purchase details, verify a record on the PhilGEPS notice, and view the data status. Viewing the price map, viewing location purchases, and comparing locations each include “Filter results,” because the system always applies the filters (item, unit price range, region/province/city, and award year) before it shows anything, and only completed projects are included.

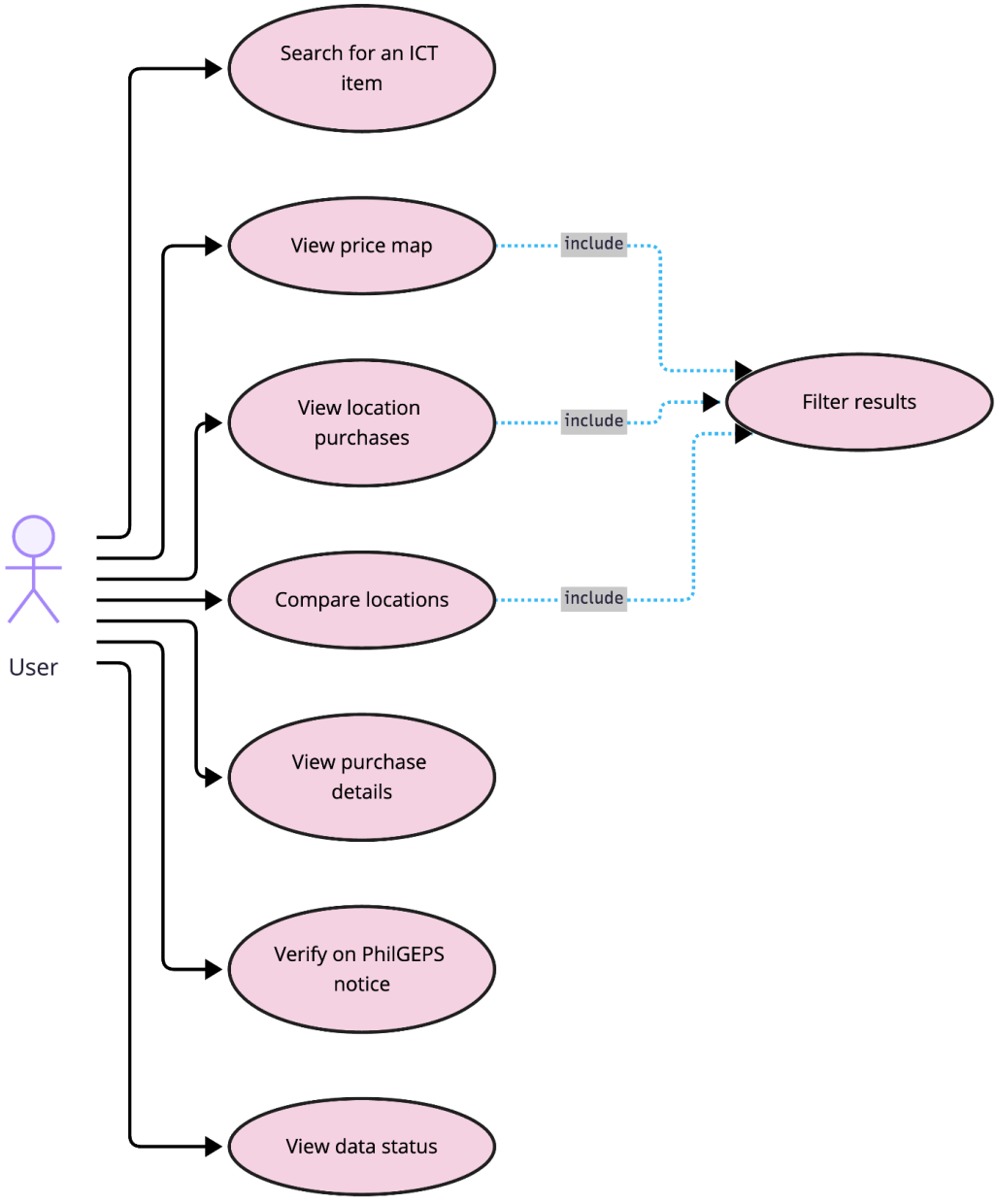


Figure 1b.2: Use case diagram for the User actor

The following table shows brief text descriptions for each use case, as seen in the previous use case diagrams:

Use Case	Description
Import PhilGEPS data	The administrator uploads a PhilGEPS data file. The system keeps completed ICT projects, groups similar items, finds each agency's coordinates, region, province, and city, saves new records without duplicates, and updates the data status.
Search for an ICT item	The user types a keyword, such as "Mac Mini," in the item search bar and picks a matching item.
Filter results	The user narrows the results by unit price range, region/province/city, and award year. Only completed projects are included. Other use cases include this one.
View price map	The system shows the selected item's prices on a map of the Philippines, with one marker per agency that has bought that specific item, color-coded from lower to higher. The user can zoom, pan, and select an agency.
View location purchases	The user selects an agency on the map and sees the list of purchases made by that agency.
Compare locations	With an item already selected, the user clicks one agency on the map. They then either click a second agency, or choose an average to compare against (the average for its region or for the whole country). The system shows the item's price at the agency, the price it is compared with, and the difference.
View purchase details	The user opens one purchase and sees the agency and location, notice title, reference number, award date, supplier, contract price, and the item description, quantity, and unit price.
Verify on PhilGEPS notice	The user clicks the link on a purchase to the original PhilGEPS notice to cross-check it.
View data status	The user sees when the data was last updated and which datasets are included.

iii. System Sequence Diagrams

As seen in Figure 1b.3, the user types an item name in the search bar, and the web page asks the system for matching items. The system looks them up in the database and returns suggestions. The user then picks the item, sets the filters (unit price range, region/province/city, and award year), and clicks the *Apply* button. The system gets the completed projects that match from the database, calculates the average price per location, and assigns each location a color group (based on price). The web page then shows the color-coded map.

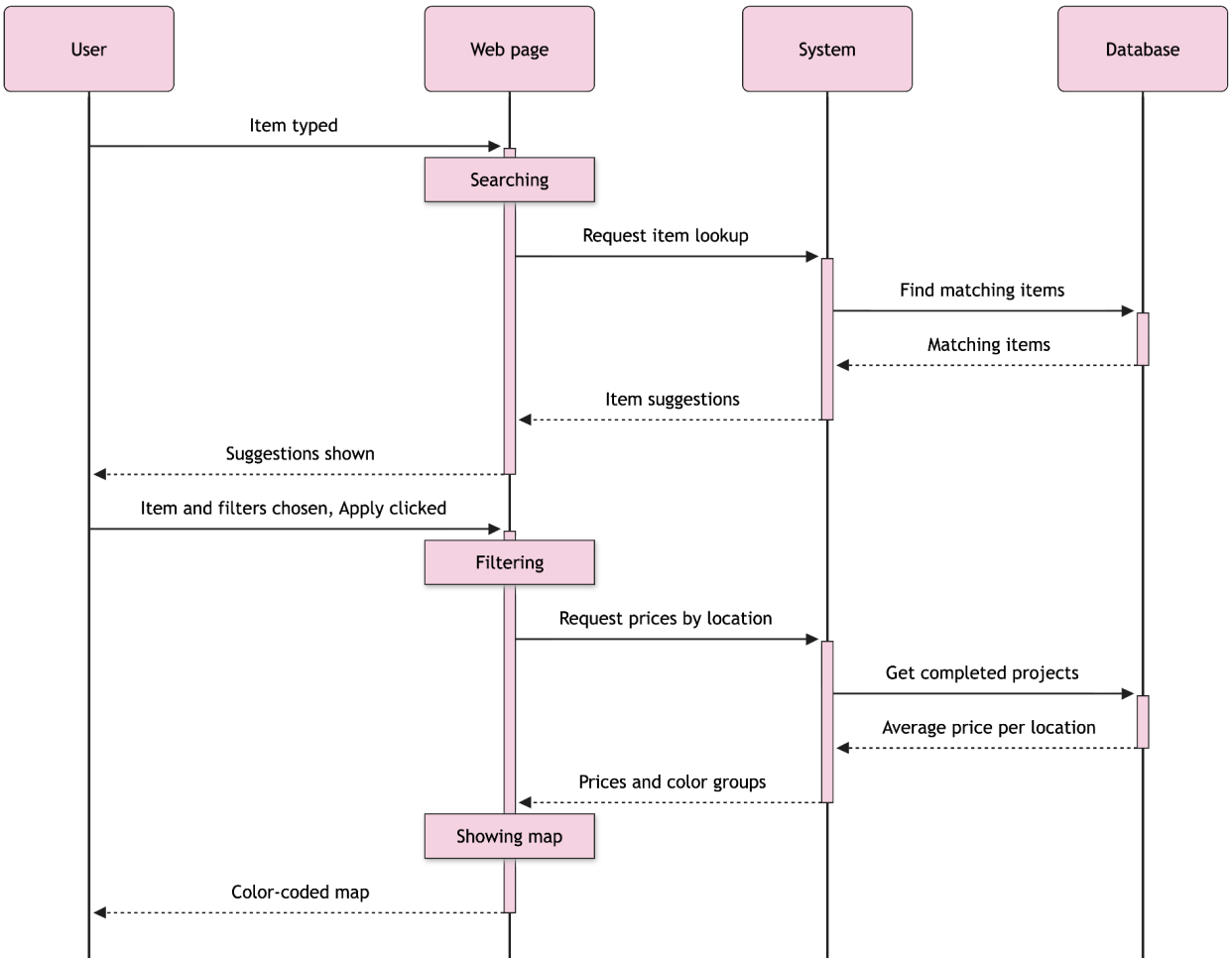


Figure 1b.3: Sequence diagram for the use cases for Search, Filter, and View Price Map

In Figure 1b.4 below, the user selects a location on the map. The system retrieves the purchases made there from the database, and the web page shows them as a list. The user selects one purchase, and the system retrieves its full record: agency and location, notice title, reference number, award date, supplier, contract price, and item description, quantity, and unit price, along with the notice link. When the user clicks that link, the original PhilGEPS notice opens in the browser so the record can be verified.

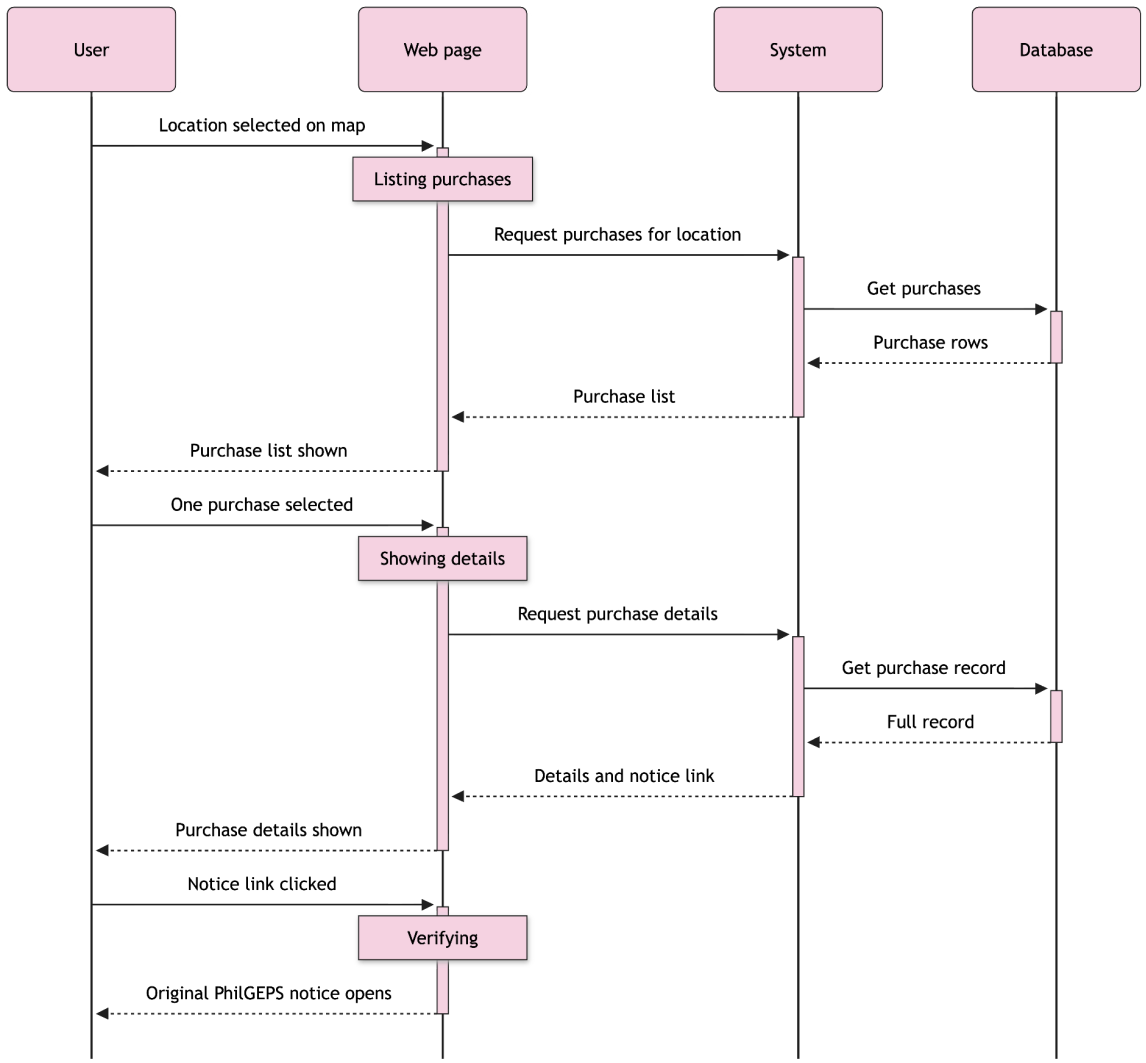


Figure 1b.4: Sequence diagram for the use cases for View Location Purchases, View Purchase Details, and Verify on PhilGEPS Notice

Figure 1b.5 shows the sequence diagram for the use case that compares locations. With an item already selected (for example, Mac Mini), the user clicks one agency on the map, such as UP Diliman, and chooses to compare. The user then picks one of two options. The first is to click another agency, like UP Manila. The second is to choose an average, like the regional or the national average for that item. The system gets the needed prices from the database and calculates the difference. The web page shows the two prices side by side with the difference.

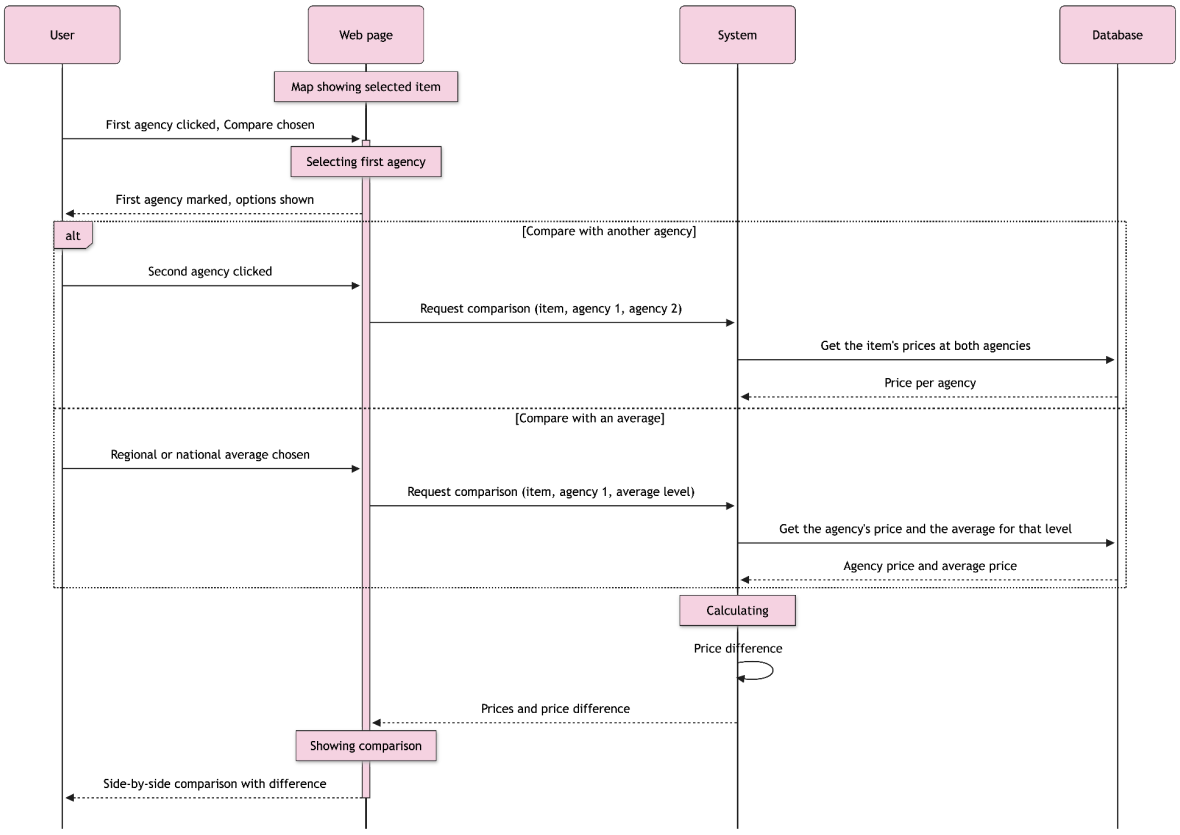


Figure 1b.5: Sequence diagram for the use case for Compare Locations

Finally, in Figure 1b.6, we see the sequence diagram for the administrator's use case where they import PhilGEPS data. The administrator uploads a PhilGEPS dataset file on the import page. The system keeps only completed ICT projects, cleans the data, groups similar items, and finds each agency's coordinates, region, province, and city. It saves the new records, skips duplicates, and updates the last imported date and the list of files included. The administrator then sees how many records were added, or if the upload skipped or failed.

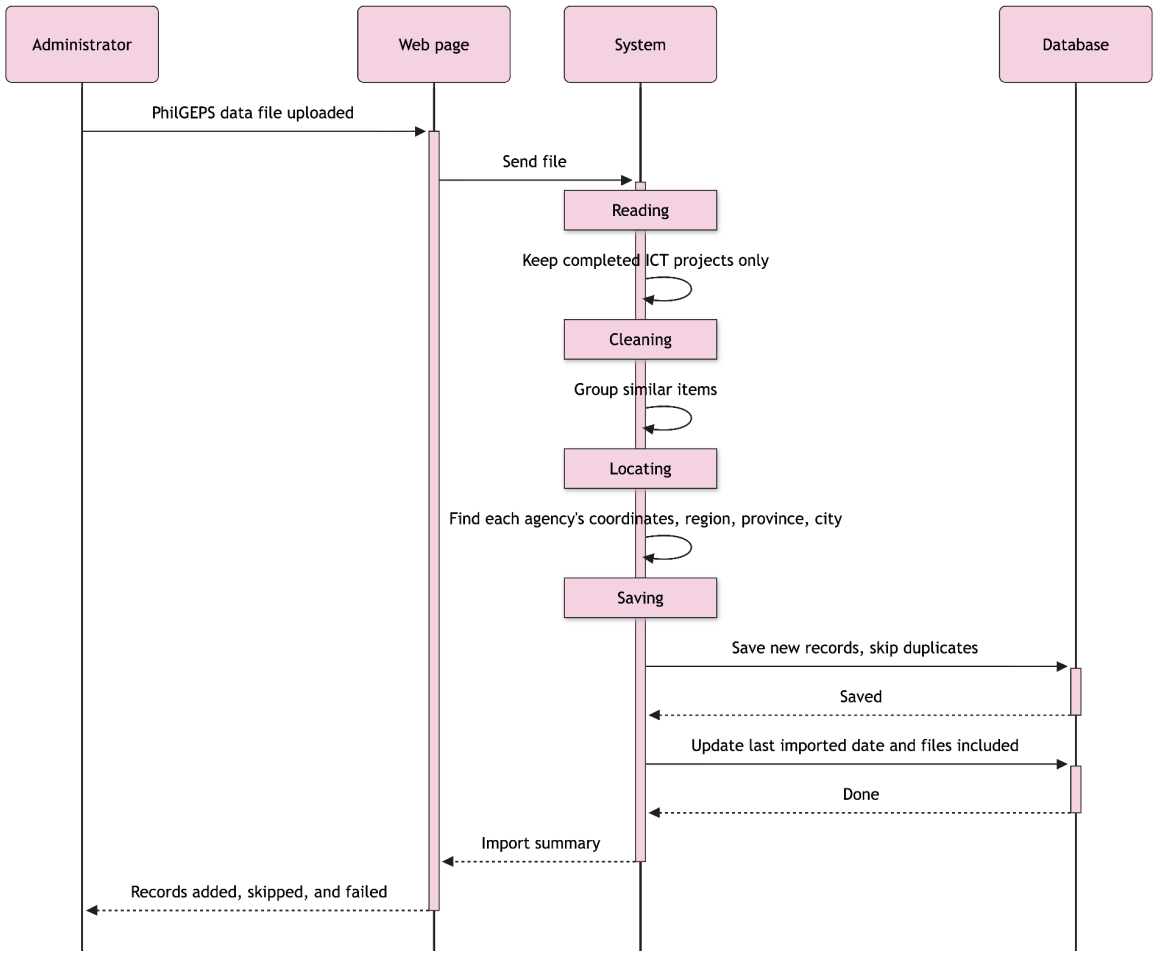


Figure 1b.6: Sequence diagram for the use case for Import PhilGEPS Data

c. Configuration Model

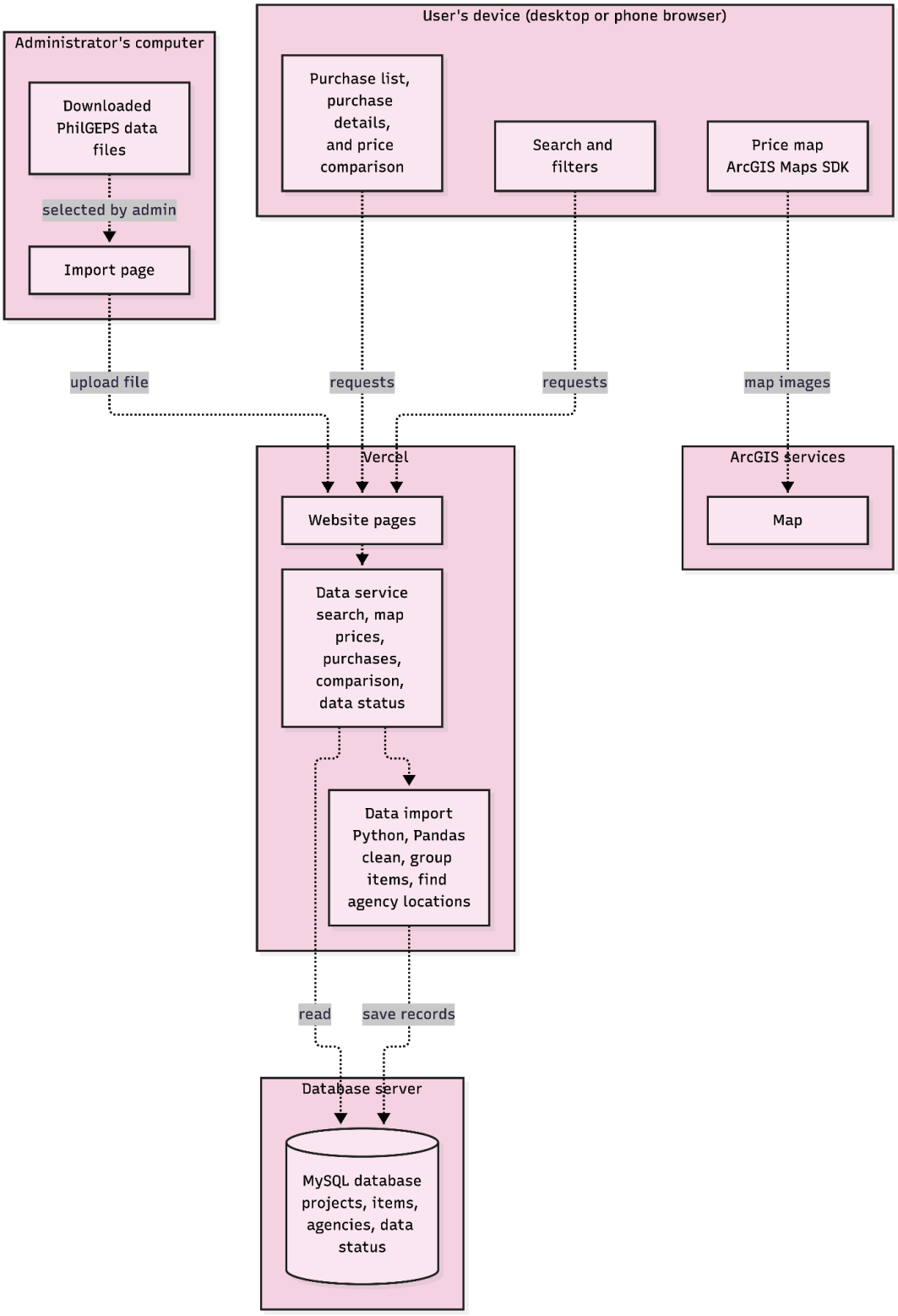


Figure 1c.1: UML deployment diagram

The system is a web application that runs on five nodes: the user's device, the administrator's computer, the Vercel host, the database server, and the ArcGIS services. Only the user's and administrator's devices are separate computers from the team's setup. The rest are hosted services.

- **User's device (desktop or phone browser)**
  This holds everything the user sees and uses. The price map uses the ArcGIS Maps SDK to draw color-coded agency markers and lets the user zoom, pan, and select an agency. The search and filters panel lets the user look for an ICT item and narrow results by unit price range, region/province/city, and award year. The purchase list, purchase details, and price comparison screens show the purchases of a selected agency, the full record of one purchase with its PhilGEPS notice link, and the comparison of an agency with another agency or with the regional or national average. This node only displays information and sends requests. It does no data processing. Phones use the same web app in their browser, so there is no separate mobile app.

- **Administrator's computer**
  The administrator downloads PhilGEPS data files and uploads them through the import page. This computer needs only a web browser.

- **Vercel**
  This hosts the server side of the app. The website pages serve the screens to the browser. The data service answers the browser's requests: it searches items, calculates the average price per agency for the map, returns purchases and purchase details, works out price comparisons, and reports the data status. The data import runs when the administrator uploads a file.

- **Database server**
  The MySQL database stores the projects, items, agencies, and the data status. The data service reads from it to answer user requests, and the data import writes new records to it.

- **ArcGIS services**
  This external service provides the background map. The user's browser fetches the map images from it directly, so they do not pass through our server.

II. Web App Architecture and Design

a. Architecture Design

The application serves two actors: the public user and the administrator. Public views provide item suggestions, filters, an ArcGIS map, purchases, details, comparisons, and data status. The administrator uploads PhilGEPS files through a restricted page.

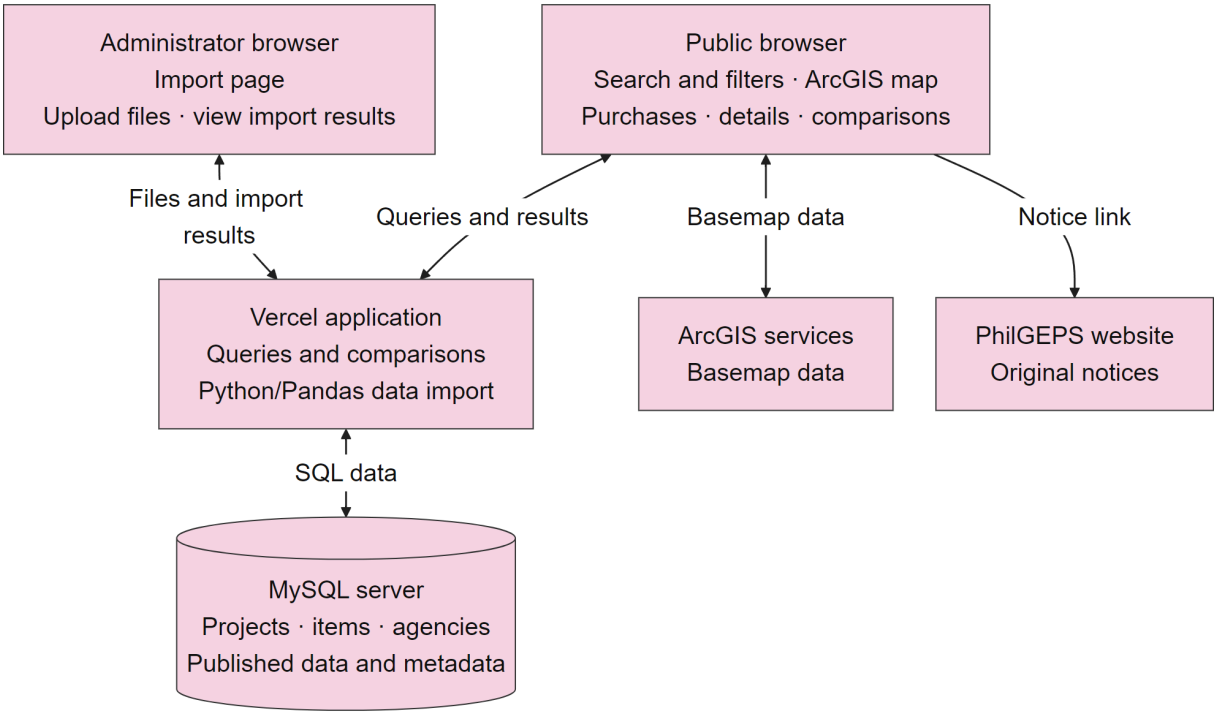


Figure 2a: Architecture design diagram

The web app uses client-server, layered, and data-centered styles. The browser presents the interface, Vercel services handle requests and business logic, and MySQL stores shared data. This separates presentation, control, and model responsibilities as an MVC adaptation. Content follows a hierarchical structure linking results, purchase details, and comparisons.

Architectural Decisions

- **Published Procurement Data**
  The application stores imported PhilGEPS records in MySQL so users can search and compare prices without retrieving source notices for every request. Data freshness depends on administrator uploads, so the interface displays the last update and

included datasets. A proposed recovery approach is to validate a cumulative dataset before activation, preserving the previous version if an import fails.

- **Server Domain Logic**
  Vercel services handle searches, price calculations, comparisons, and data imports using Python/Pandas. The browser displays results and retrieves basemaps directly from ArcGIS. Centralizing rules keeps calculations consistent, while availability depends on the hosting, database, and map services.

- **Comparable Item Prices**
  Comparisons use completed ICT purchases with compatible specifications, units, and supported unit prices. Budgets and mixed contract totals are excluded from unit-price calculations. This reduces misleading comparisons, but records with insufficient information cannot contribute to the averages

Quality Requirements and Design Responses

Category	Quality Attribute	Design Response and Acceptance Check
Usability	Global site understandability	Stable price map, data status, and help entries.
	Feedback and help	Show loading, empty/error states, and import summaries beside the affected work. Retain context when opening help.
	Interface and aesthetics	Consistent controls, responsive panels, a readable price legend, and aligned PHP amounts.
Functionality	Search and retrieval	Item suggestions and selection precede the unit-price, geography, and award-year filters.
	Navigation and browsing	Restore filters, dataset version, and map/list position when returning from purchase details.
	Domain features	Compare what one agency paid for the selected item with another agency's price or another domain's average.
Reliability	Correct links	Preserve source identifiers and original notice URLs.

	Error recovery	Validate an uploaded batch before activation. Failed import leaves the preceding published data usable.
	Input validation and recovery	Reject invalid ranges, IDs, years, and files. Explain corrections without clearing other entries.
Efficiency	Response time	Indexed queries and bounded responses.
	Page generation	Return a small page shell and compact results. Measure page and query timings separately.
	Graphics generation	Render one summary marker per matching agency. Test ArcGIS rendering with the expected marker count.
Maintainability	Ease of correction	Separate source parsing, validation, database access, calculations, and presentation.
	Adaptability	Version source column mappings and comparable item rules when source formats change.
	Extensibility	Store administrative hierarchy and dataset coverage as data so later regions and years can be added.

b. Content (Data) Design

i. Class Diagram

This class diagram models the core data structures and service components of the GIS for ICT Procurement web app, capturing relationships between items, agencies, procurement contracts, filters, and comparisons.



Figure 2b.1: Data structure and components class diagram

ii. Entity-Relationship Diagrams

This diagram defines the core database schema for the system, modeling relationships between government agencies, standardized items, procurement projects, and data tracking statuses.

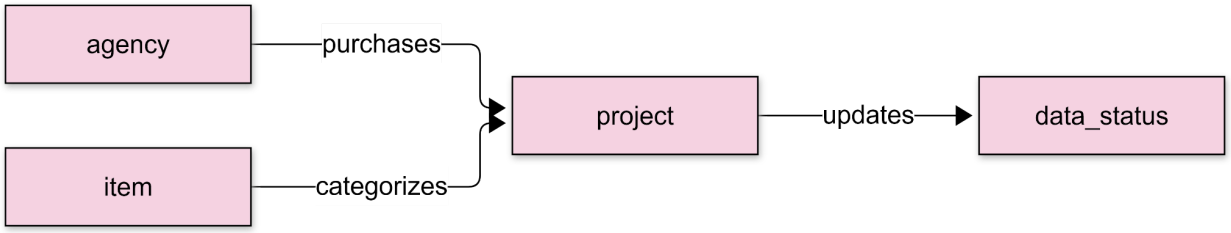


Figure 2b.2: Entity-Relationship diagram

iii. Data Flow Diagrams

These DFD diagrams (from level 0 to 2) illustrate the hierarchical transformation of data within the application, starting from the high-level context boundary down to internal subsystem processes like CSV data ingestion, keyword parsing, and spatial map rendering.



Figure 2b.3i: DFD level 0

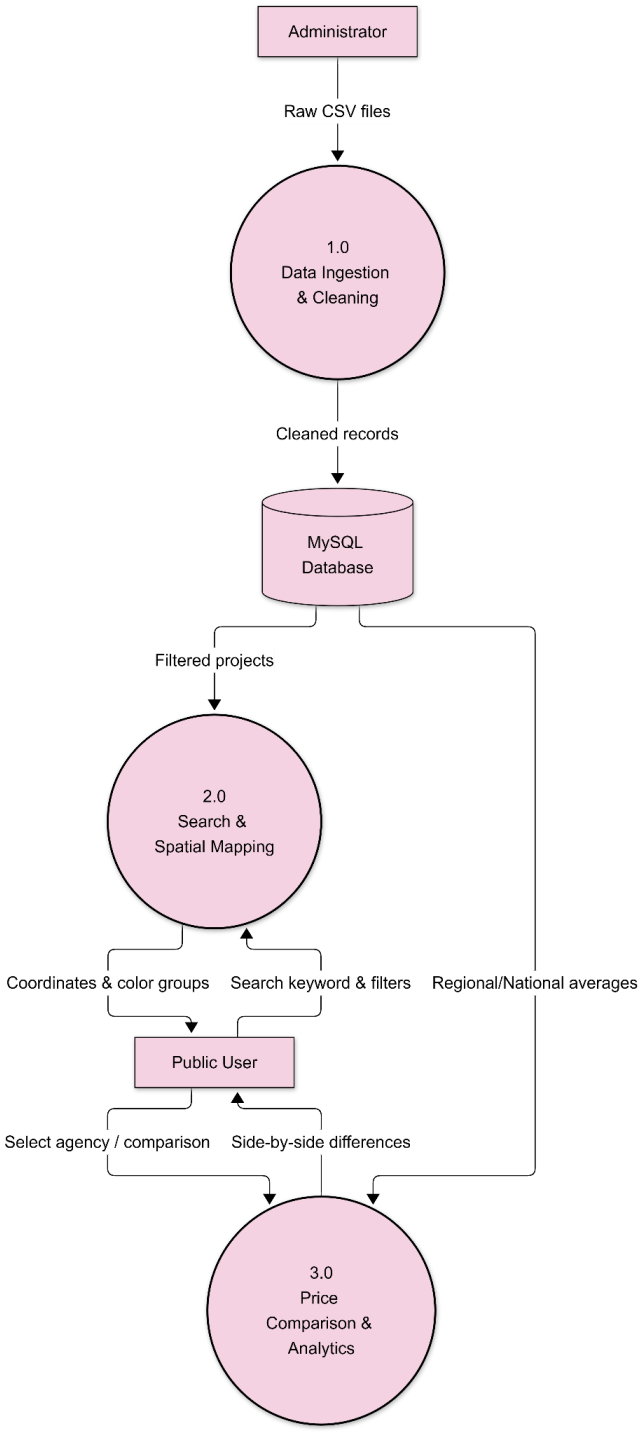


Figure 2b.3ii: DFD level 1

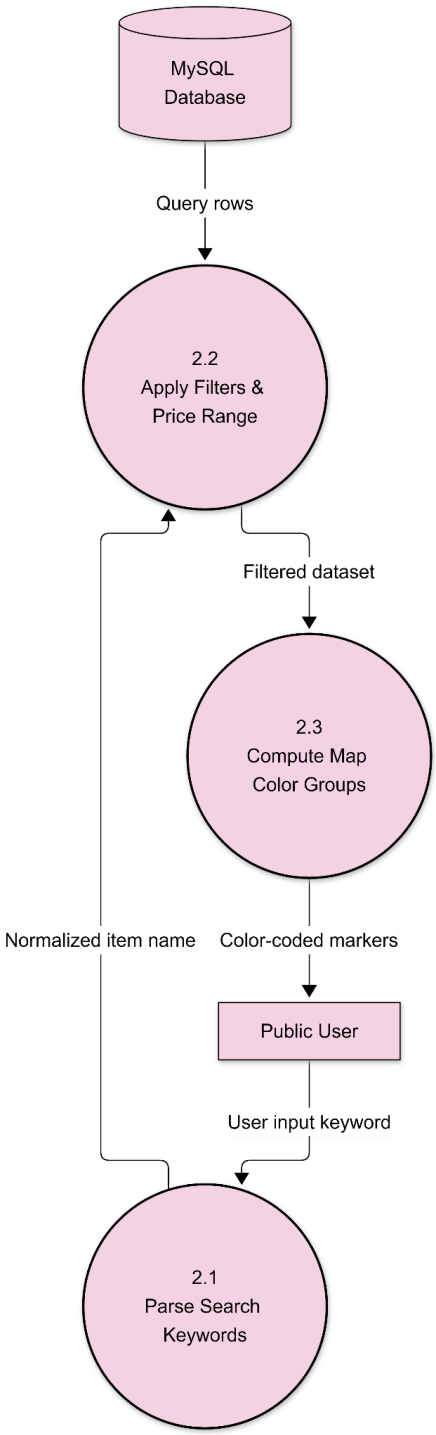


Figure 2b.3iii: DFD level 2

c. Component-level design

i. Algorithms

Data Ingestion and Item Category Classification

The data ingestion and classification algorithm processes raw PhilGEPS CSV uploads through a structured, multi-step pipeline using Python and Pandas. The procedure begins by reading row data and filtering out unawarded, cancelled, or failed bids to retain **strictly completed ICT procurement projects.**

For each valid record, the algorithm extracts free-text item descriptions and maps them to standardized **UNSPSC** codes to unify disparate naming conventions. It then performs an agency geocoding lookup to assign precise latitude and longitude coordinates, checks for duplicate entries to maintain data integrity, and saves the normalized records directly into the MySQL relational database before updating the system's tracking metadata.

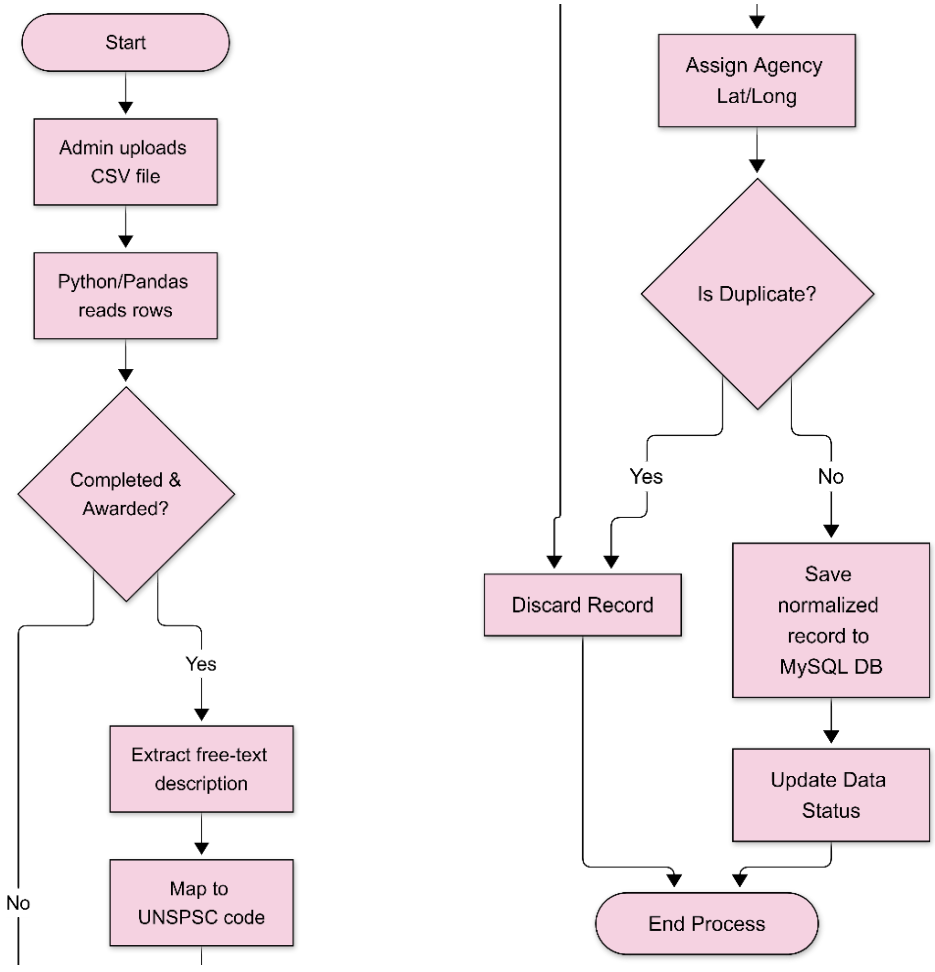


Figure 2c.1a: UML activity diagram for the data ingestion and item classification algo

Price Comparison and Color-Coding Algorithm

The price comparison and color-coding algorithm evaluates procurement costs dynamically when a user queries a specific ICT item. Upon receiving the search parameters, the backend queries the database to retrieve matching records and computes regional and national average unit prices.

The algorithm then compares individual agency unit costs against these benchmark averages, categorizing them into distinct pricing tiers (we assign **green** markers for **lower-cost** outliers, **yellow** for **standard ranges**, and **red** for **higher-cost** acquisitions). Finally, it renders these color-coded markers on the GIS map interface alongside summarized statistical analytics.

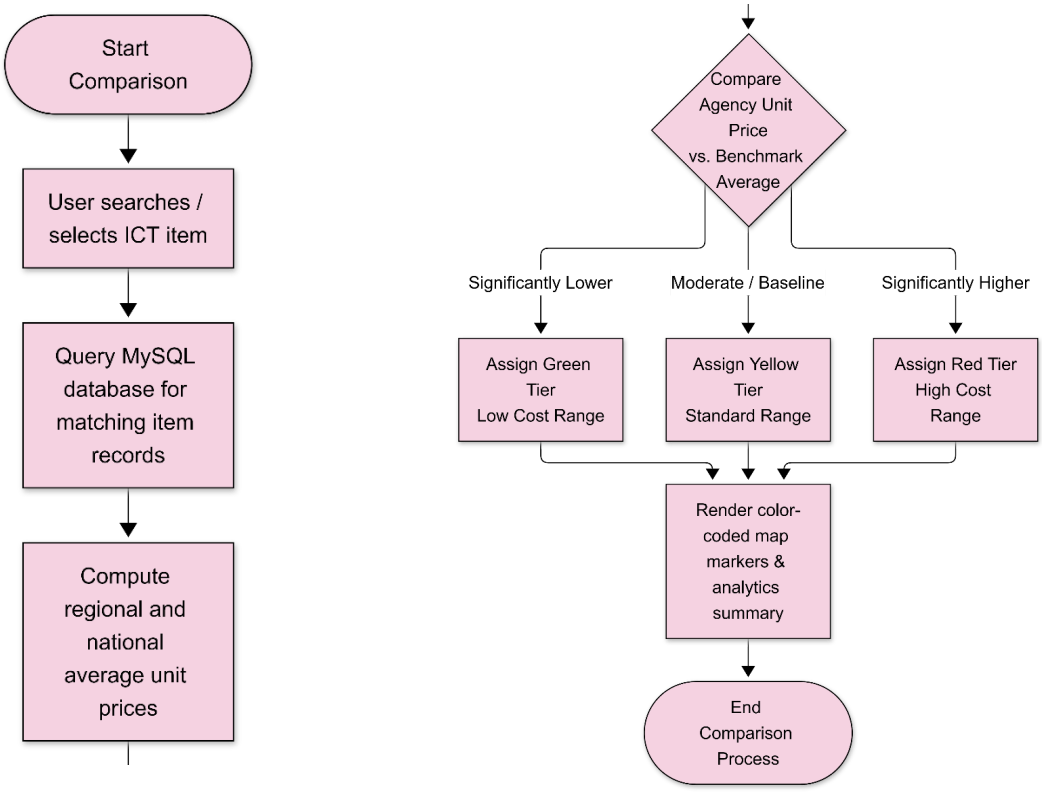


Figure 2c.1b: UML activity diagram for the price comparison and color coding algo

ii. Data Structures

Data Structure	Implementation	Purpose	Why
Relational Tables	MySQL Database	Persistent storage for agency, item, project, and data_status entities.	*For relational integrity and fast multi-variable filtering across prices, locations, and dates.*
Hash Maps / Dictionaries	Python / In-Memory Backend	O(1) lookups during CSV ingestion to map raw strings to UNSPSC codes and coordinates.	*For execution speed to resolve classification matching.*
GeoJSON Arrays	Backend API & Frontend	Lightweight spatial data arrays packaging latitude, longitude, and pricing tiers for map rendering.	*For compatibility with ArcGIS mapping SDKs, and faster rendering.*

d. Interface elements / Usability design

The interface supports the two actors in Part I. Public users search and compare completed ICT purchases, while the administrator imports PhilGEPS files through a restricted page. Each actor has a separate screen and task flow with consistent labels, feedback, and help. These layouts map user goals to controls.

i. Interface Design - Menus and links

Public User Interface

Figure 2d.1a shows the general arrangement of the public interface. The header is at the top, search and filters are on the left, the map occupies the center, and a summary appears on the right. Related purchase information is grouped below the map.

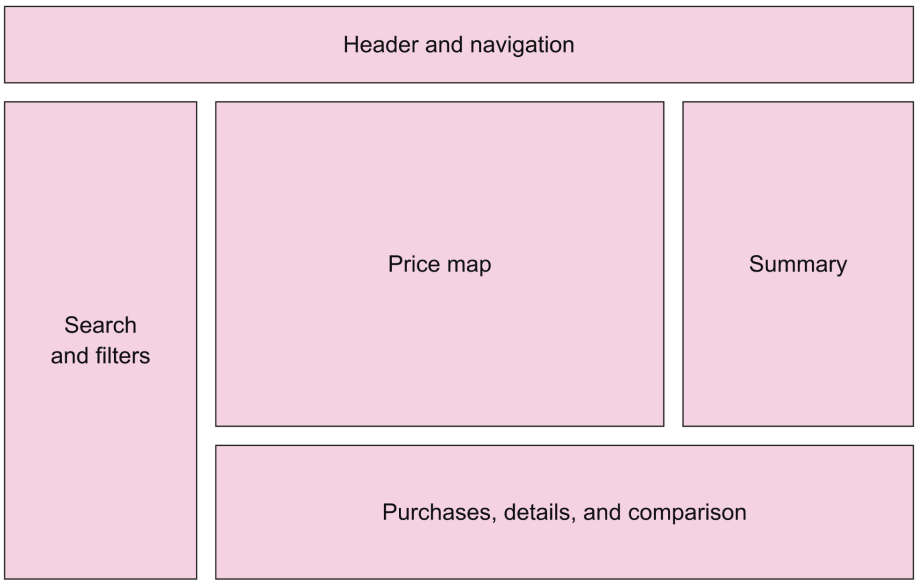


Figure 2d.1a: Interface layout for public user

The search and filter area lets users narrow the records shown on the map. The summary presents information for the current selection. The lower area provides access to purchases , details, and comparisons. These modules support the public-user tasks described in part i while keeping related information together.

Administrator Interface

Figure 2d.1b shows the administrator dashboard. The system head is at the top, and a navigation menu on the left provides access to import data and data status. The main workspace contains the upload area, import results, and data status. Access is restricted to authorized administrators.

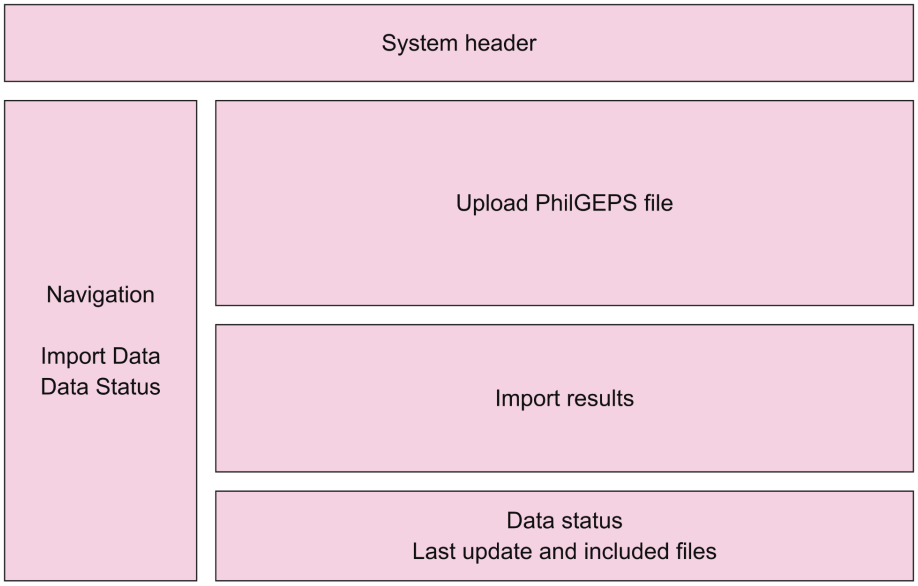


Figure 2d.1b: Interface layout for administrator

The upload area occupies the upper part of the main workspace, where the administrator selects and submits a PhilGEPS file. Import results appear directly below it, showing progress and the numbers of records added, skipped, or failed. This keeps the main import task and its outcome together.

The results area explains errors so the administrator can correct the file and try again. The read-only data status section at the bottom shows the last successful update and included files. The left menu gives access to this import-related information.

ii. Aesthetic Design - Layout and graphics

Both layouts use clearly separated sections with short headings and consistent text. The public interface emphasizes the map, with supporting information beside and below it. The administrator dashboard places navigation on the left and the import workspace on the right. Related controls and information are grouped within each section.

The following table addresses the interface-design issues.

Design Issue	Planned Interface Behavior
System Response Time	Show progress, prevent duplicate submissions, and offer retry after failure. Evaluate response time and its variation.
Help Facilities	Provide task-specific help and a clear return action that preserves selections.
Error Information	Identify the problem, affected field or row, recovery action, and whether data changed. Retain valid entries.
Command Labels	Keep action labels consistent: Apply, Clear filters, View purchases, View details, Compare, Choose file, and Import.
Accessibility	Provide keyboard access, visible focus, labeled controls, readable contrast, and usable touch targets. Keyboard selection of an agency opens its purchase list.
Internationalization	Preserve Unicode names. Use PHP amounts, clear dates, and Asia/Manila update times.

iii. Navigation Design

Figure 2d.2 groups related tasks into navigation semantic units. The user and administrator actors identify each pathway's entry point. Labeled arrows connect search, purchases, details, and comparisons, with a specific import pathway for the administrator.

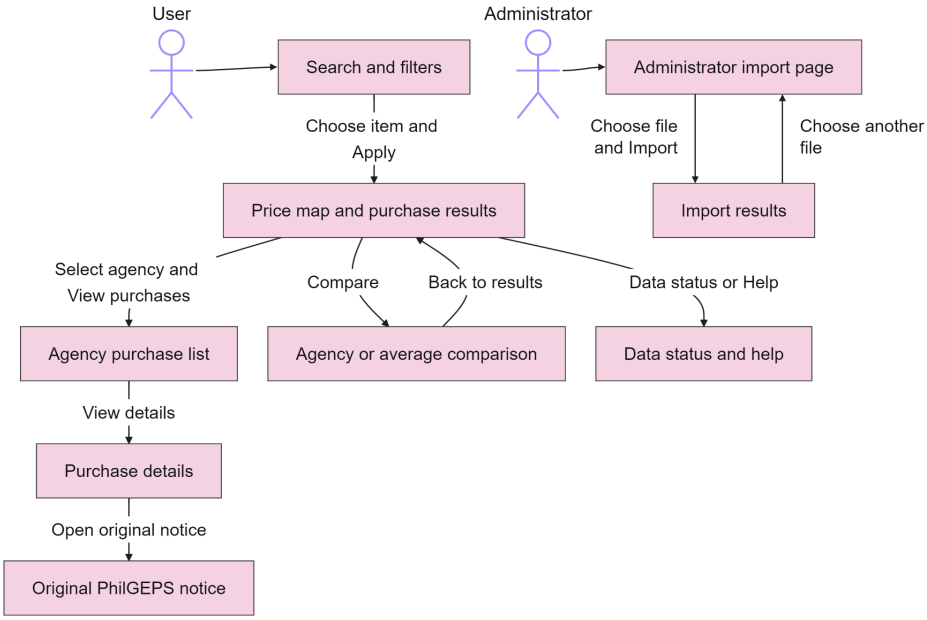


Figure 2d.2: Navigation semantic units for users and administrator

Returning restores the selected item, filters, and results. PhilGEPS links open in a new tab. Data status and help remain accessible. Failed requests offer retry, while import results allow a return to file selection.

Navigation Task Estimates

Each click or typed character counts as one action. Data entry includes typing and selecting input values or files. Navigation includes moving between controls, opening views, and submitting commands. These estimates assume a loaded desktop page and exclude login, scrolling, and loading time.

Task and Starting Point	Action	Navigation	Data Entry	Total
Find purchases, starting at the search screen	Select the search field; enter an ICT item keyword; choose a matching suggestion; Apply; select an agency; view its purchases	4 (31%)	9 (69%)	13

Verify a purchase, starting at its agency's purchase list	View details; open the original PhilGEPS notice	2 (100%)	0 (0%)	2
Compare agencies, with the first agency already selected	Compare; select the second agency	2 (100%)	0 (0%)	2
Import data, starting at the administrator's import page	Choose file; select the file; Open; Import	3 (75%)	1 (25%)	4

The search estimate assumes an eight-character keyword. The import estimate assumes the file is immediately visible in the file picker. Longer keywords, additional filters, or file searches increase the action count. These estimates will be checked during prototype testing.